

J! ROYAL RUMBLE

DESIGN AND RULES

A thirty-player elimination trivia format built on the Jeopardy! clue structure and the entry mechanic of the 1980s Royal Rumble. This document covers the rules in full, the optional mechanics, and the design work underneath — including the two problems that nearly broke the format and what the numbers said about fixing them.

Every figure here comes from the simulator or from a recorded match. None of it is illustrative.

THE SHAPE OF IT

Three players start. The rest wait in draw order and enter one at a time as the match runs. Answer a clue correctly and every other player in the ring pays you its value. Miss and you pay, and the clue goes back out to everyone else. Fall below zero and you are gone. The last player standing wins, and nothing else counts as a result.

A full field takes about half an hour. The board is six Jeopardy! categories at a time, drawn from an archive of 47,000 categories or from boards written in house, in whatever ratio the host sets.

The one number that matters

Under the obvious scoring rule — a correct answer is worth the clue value, the way it works on television — the player who drew the last entry number won **100%** of simulated matches. Not most. All of them. Everything in the section on scoring exists because of that number.

THE RULES

The board

Six categories are live at once, five clues each, \$100 to \$500 by row. Row position sets the value regardless of where the category came from — a category lifted from a Double Jeopardy round plays at the same values as any other, and is simply harder. When a category is used up it is replaced immediately, so the board never runs out.

The host can reroll any category before or during play. Three rerolls of the same category retires it permanently.

Entering

Every player draws an entry number before the match. Numbers one to three open the match; the rest enter on a fixed clue interval, which the host sets or lets the app scale to the size of the field. Each entrant walks in with the same stake.

Scoring

Correct. Every other player in the ring pays you the clue value. You collect all of it. With five in the ring a \$500 clue is \$2,000 to you and \$500 from each of the other four; heads-up it is worth \$500.

Wrong. You lose the value and you are locked out of that clue. Nobody else is affected by your miss.

The re-toss. A missed clue goes back out as a genuinely fresh race: the buzzers reopen for everybody still eligible, including anyone who did not attempt it the first time. If somebody else converts it you pay them along with everyone else, so a miss followed by a conversion costs twice.

Nobody gets it. Every player in the ring loses half the value. Passing is not free.

Elimination

Below zero and you are out; exactly zero survives. If one answer sinks several players they all go together, and the player whose answer did it is credited with a **toss out** for each. If a clue would wipe the entire ring, the highest score survives.

The ceiling

There is a maximum score, and it falls across the match. Anything above it is clipped. It never drops below the amount a new entrant carries in, so nobody arrives already capped.

Clearing the field

Eliminate everyone in the ring while players are still queued and you take a bonus equal to every clue left on the board. The board is replaced entirely and two fresh players enter rather than one.

Buzzing

The host reads the clue, then arms the buzzers. Fastest reaction takes it, one buzz per player per race.

Buzzing before the lights costs a 250 millisecond lockout, and the penalty runs even if the buzzers open while it is being served.

Players still in the queue, and players already eliminated, can buzz on every clue. It does not score and cannot affect the match, but it reports their time and where they would have placed against the live field. Arriving with your timing already calibrated is a real advantage.

Times under 150 milliseconds are normal

Strong players do not react to the lights. They learn the rhythm of the host's read and time the buzzer to land the instant it opens. A perfectly judged buzz reads 0.0. In the first recorded match, the two active players buzzed a median of 197 and 207 milliseconds with best times of 8 and 17 — and one of them jumped the lights on a third of his presses. That is the trade the lockout exists to price.

WHY THE SCORING WORKS THIS WAY

The draw-position problem

The natural rule is the televised one: a correct answer is worth the clue value to the player who gets it, and the format supplies the drain by taking that value off everyone else. Simulated across four thousand thirty-player matches, that rule produced this:

Draw numbers	Share of wins	Fair share
1 – 10	0.0%	33.3%
11 – 20	0.0%	33.3%
21 – 25	3.1%	16.7%
26 – 30	96.9%	16.7%
— of which draw 30 alone	46.7%	3.3%

Four thousand simulated matches. Draws 1 through 21 did not win once.

The cause is arithmetic rather than luck. Under flat scoring a player's expected change per clue is $V(2 - P)/P$, where V is the clue value and P the number of players in the ring. For any P above two that is negative for everybody. Nobody can accumulate; the whole field drifts downward at roughly the same rate. Across an entire thirty-player match the highest score anyone reached was around 5,950 against a 5,000 starting stake — so the score ceiling never bound, the field-clear bonus almost never triggered, and a fresh entrant arriving on 5,000 simply outlasted incumbents who had been grinding down for half an hour.

The fix: collect from each opponent

The loss side is left alone — it is what makes the format an elimination game. The gain side changes: the answerer collects the clue value from every opponent rather than once. Elimination pace is untouched, but a strong player can now build a wall, which is what allows an early entrant to survive long enough to face the last numbers.

It is also the more natural sentence to say out loud at the table: *everyone else pays you*.

The ceiling, and why it falls

Letting players accumulate reintroduces a different problem: a runaway leader. A fixed ceiling solves that but hands the advantage straight back to the late draws, because it clamps exactly the players who have been building. A *rising* ceiling is worse still — it is tight while the early entrants need room and generous by the time the last numbers arrive.

Ceiling	Match length	Back-half draws win
Fixed 6,000	67 min	72%
Fixed 10,000	118 min	55%
Rising, 4,000 +40/clue	100 min	73%
Falling, 15,000 –50/clue	76 min	53%

The tuning as it stood at the time. 50% is fair. This is the reasoning that was later overturned — see the next section.

That was the case for a decaying ceiling, and it held for months. It is wrong, and the section that follows explains why. The floor is still the entry stake, so a late arrival is never capped on the way in.

What the tuning produced

Field	Entry every	Start	Ceiling	Decay	Length	Back-half
10	10 clues	3,000	7,500	–40	35 min	57%
16	7 clues	3,000	7,500	–25	46 min	62%
20	6 clues	3,000	7,500	–25	48 min	64%
30	5 clues	3,000	11,000	–40	60 min	63%

Back-half is the win rate of players drawing in the second half of the field. It started at 100%.

A late number is still worth having, and that is deliberate — it is true to the source material, where entrants 27 through 30 genuinely do overperform. What it is no longer is a guarantee.

OVERTIME

Two evenly matched players trade the same points back and forth indefinitely. Heads-up, a correct answer is worth exactly what the opponent loses, so a field of two with similar ability is a random walk with no drift. It showed up in the first recorded match: the last fourteen clues oscillated without either player getting meaningfully closer to going out.

Once the queue is empty and the ring is down to two, clue values double every six clues, capped at eight times face. The board displays the raised values, and the room gets a full-screen announcement at each step.

Endgame	Result
Perfectly alternating stall, no overtime	Still running at 400 clues
Perfectly alternating stall, with overtime	Resolved in 22 clues

The pathological case, since real players are never quite this symmetrical.

THE CEILING, AND WHY IT STOPPED FALLING

The ceiling clips every score. It began falling across the whole match, to stop a late draw walking into a worn-down field and simply outlasting it. That reasoning was backwards.

The ceiling clips whoever is ahead, and whoever is ahead is nearly always an early entrant who has been accumulating. A latecomer arrives at a fixed stake, untouched. So the falling ceiling was destroying exactly the work it was meant to protect against.

Decay per clue	Last third vs first third
-180	9.0×
-40	2.16×
-25	1.85×
0	1.40×

3,000 simulated twenty-player matches per row. 1.00 would be even.

A measurement that reversed the answer

The first version of this simulation used a random comparator to shuffle the buzz order, which is not a shuffle at all — it leaves a heavy bias toward the original order. It reported that the first three draws, who all start together, won at 25.8%, 14.2% and 10.4%. Three players in identical positions cannot differ like that. With a proper shuffle they came out within noise of each other, and the recommendation flipped.

What replaced it

Taking the decay out removed the only guaranteed drain, and a symmetric exchange with nothing leaking never resolves — a field of evenly matched robots ran 400 clues without a single elimination. Raising the stakes does not help, because doubling both sides of an even trade leaves it even.

So the ceiling falls only once overtime has opened. The endgame gets teeth; the entry phase, where the bias came from, is left alone.

PAID FOR LASTING

Early draws spend the whole match being ground down. Rather than take from the leaders, the game now pays for survival.

Bonus	10 players	20 players	30 players
none	1.31×	1.40×	1.14×
every 10, +250	1.18×	1.28×	1.07×

every 10, +500	1.05×	1.07×	0.97×
every 10, +1000	0.82×	0.89×	0.80×

Draw advantage of the last third over the first. +1000 overshoots and hands the game to early draws instead.

It self-limits: a leader at the ceiling gets nothing from it, so it helps whoever is grinding rather than whoever is already winning. Taking every clue in a column pays another 500, scaled by the overtime multiplier, and is counted as the column is worked through so a board refresh cannot rob anybody of a run they have finished.

OPTIONAL MECHANICS

Four rules that change how the game is played rather than how it looks. All off by default, each toggled separately, and all of them lean on keyboard controls — worth leaving off if the field is mostly on phones.

Top rope

Declared between clues, never once a clue is on the board — otherwise you would only ever climb up when you already knew the answer. That clue is worth double in both directions to the player who declared, and their winnings ignore the ceiling. That last part matters: without it the top rope is strictly bad for anyone near the cap, since the downside is uncapped and the upside is not.

Targeting

Aim at one player, visible to the room, with an alert on their buzzer. Win the clue and the entire pot comes out of them alone and everyone else is spared. Lose the clue to them and you pay the whole pot yourself while the rest of the ring walks. A finishing move and a kamikaze on the same button.

Bounties

A player waiting in the queue stakes part of their own entry on a head, up to half. Whoever eliminates the target collects. If the target survives to the end they keep it — and if the target eliminates the player who placed it, they keep that too. Placing a bounty is a declaration, not a free shot.

Revival

An eliminated player returns to the queue at a fraction of the stake with a new entry number, once by default.

Setting	Length	Back-half draws win
None (baseline)	64 min	53%
Top rope	62 min	54%
Targeting	63 min	63%
Bounties	64 min	48%
Revival	95 min	52%
All four	88 min	63%

400 simulated thirty-player matches per row.

Three of the four barely move the clock. Revival runs half again as long, because nearly every player spends their second life — the setup screen says so before the match starts. It is also the fairest setting on the list, which was not the expectation: a second chance is worth most to whoever went in first, so it works against the same late-draw advantage the longevity bonus exists to fight.

A measurement trap worth recording

Revival first showed a 94% back-half win rate, which looked alarming. It was an artefact: revived players receive a fresh entry number, so the winner is a late draw almost by definition. Measured against the number each player *actually drew*, it is 52%. Players now carry both numbers and the standings report the drawn one.

RUNNING A MATCH

Before

The setup screen mints a four-letter room code to read aloud, and fills a lobby as players join. Clue material is mixed by percentage across the archive, boards written in house, and anything uploaded for that match; the archive can be narrowed by season. Fresh boards can be dropped in as j-trivia.org JSON or jparty.tv CSV.

The screen estimates the length from the roster and warns when the numbers do not work — a six-player field cannot be stretched much past its natural length by entry pacing alone, and it says so, with a button that sets a target it can actually reach.

During

- The clue takes over the shared screen. Correct responses appear only in a separate admin window, kept on an unshared display.
- Three keys resolve everything: correct, wrong with automatic re-toss, and nobody got it.
- A five-second lectern countdown runs after the buzzers arm.
- Undo takes back an entire clue — scores, eliminations, entries, the board. Clicking any player adjusts their score directly. Both are logged.
- The audio delay is adjustable mid-match, because nobody can tell you it is wrong until they have tried to buzz on a real clue.

Robots and testing

- Add robots from the setup screen, mixed across the five standards or all of one.
- Click any robot to change its standard, or remove it, before the match starts.
- Their buzz distributions come from real recorded play rather than a fitted curve.

Timing

The sockets are much faster than the call audio, so buzzers are deliberately held back to arm when the host's voice actually arrives. Measured on a real match with players spread across the country:

Path	One way
Socket, fastest player	7 ms
Socket, median player	14 ms
Socket, slowest player	29 ms
Zoom audio, typical	150 – 300 ms

Socket figures from a recorded match; the spread within each player was only a few milliseconds. Zoom is the industry-typical range.

So the socket beats the voice by roughly 120 to 290 milliseconds, and that gap is what the delay setting closes. Players also get their own trim in ten millisecond steps, since audio paths differ by client, buffer, and whether somebody is on headphones.

After

The winner gets the screen. Everything else — clues survived, toss outs, points drained, peak score, buzzer attempts, win rate, average and best times, the single fastest buzz of the match — goes into sortable standings and a score graph, shareable as a summary card, a full stats sheet, or one wide CSV.

With recording switched on, the match also produces a detailed log: every clue, every buzz time and the connection it arrived over, scores before and after, entries, eliminations, corrections, and a comparison of the predicted length against the real one. That file is how the model gets better.

What a recording changed

The first recorded match produced a median pace of 15.0 seconds per clue against a mean of 21.2 — six discussion breaks between 30 and 57 seconds dragged the average. It also exposed two broken statistics and an endgame that could not end. Three rules and four bugs came out of one file.

ROBOT PLAYERS

Thirty humans are hard to assemble for a test. Robots fill a roster and play the match themselves — buzzing on a real clock, with reaction times drawn per clue, so the race fills in the way it does with people. The host sees whether each one is about to be right, since there is no way to adjudicate a machine.

Five standards, drawn at random or forced, and any of them can be changed by clicking the robot before the match starts.

Standard	Attempts	Accuracy	How often drawn
rookie	23 – 35%	70 – 80%	5%
normie	37 – 61%	77 – 87%	60%
champ	63 – 70%	79 – 89%	23%

superchamp	72 – 88%	83 – 90%	11%
elite	89 – 96%	87 – 95%	1%

Attempts as a share of the clues played. These are Matt Schiffler's figures from the generator he wrote for his own game, not a reconstruction of them.

Checked against 3,339 real player-games from J!ometry's published box data, his intuition holds up unusually well. The share of real games falling in each of his bands runs 3.1%, 55.8%, 17.5%, 12.8% and 0.1% — against the 5, 60, 23, 11 and 1 per cent at which he draws them. His population's mean attempt rate is 56% against a real median of 57%.

Most players are ordinary

The weighting matters more than the bands do. An earlier version of this drew the five standards uniformly, so one robot in five was elite. On the show it is one in a hundred. A test field of twenty per cent elites is not a test of anything.

Difficulty by row

A player's attempt rate is raised to a power per row rather than multiplied by one: $\text{rate}^{\text{exponent}}$, with exponents of 0.48, 0.67, 0.85, 1.10 and 1.40 from the cheapest row to the dearest. The power form grades itself, because a fraction raised to a power above one falls away much faster when the fraction is small.

Base attempt rate	On the dearest row	Lost
30%	13%	17 points
60%	42%	18 points
90%	84%	6 points

A weak player loses far more to a hard clue than a strong one, and no per-standard table is needed to produce that. In play a rookie goes for the cheapest row 3.6 times as often as the dearest; an elite 1.1 times.

What the real data changed

The same file splits each game into its two rounds, and Double Jeopardy clues average four times the value of Single Jeopardy ones. Between the rounds, accuracy falls 1.9 points while attempts fall 15 per cent.

Difficulty lives in the attempt, not the answer

A hard clue does not make a player wrong. It makes them not buzz at all. Earlier versions of this model spread accuracy eighteen points across the five rows; the real figure is about two. Weak players are quiet rather than mistaken, which is a different thing to play against and a more accurate one.

Grouped by contestant across 1,772 people, the ladder from one appearance to ten or more runs 46.3% to 56.3% on winning the buzz, and 81.8% to 89.3% on accuracy. Both matter. The buzzer matters more.

Levelling robots to the room

The recorded robots were measured against a player whose median buzz was 43 milliseconds. Dropped in front of somebody buzzing at 400 they would be unbeatable, so they are shifted to sit alongside whoever actually turned up — measured once, after six human buzzes, and then frozen.

Two ways to get this wrong, both of them found in play

Recomputing the shift every clue made the robots chase the human: slow buzzes early dragged the field down permanently, and by the end of one match the player had sped up to 294ms while the robots sat at 612. Freezing it instead introduced the opposite fault — the human was eliminated at clue 12 having buzzed seven times, so the calibration never fired and he lost every race to robots that had never been levelled to him. It now settles after six buzzes and starts from a 190ms default rather than from nothing.

OVERTIME, AS IT ENDED UP

The rule that forces a resolution took three attempts, and each failure is worth recording because each looked reasonable at the time.

It began by waiting for heads-up. A robot test then ran thirty clues with three players trading the same points and it never fired — so it now opens as soon as the queue is empty.

Opening it there and letting it run regardless made large fields a lottery: a thirty-player match had the stakes doubling with fourteen still in the ring, and the strongest players' win rate fell from 42% to 34%. So the escalation clock counts only clues where nobody was eliminated. A stall is precisely a run of clues with nobody going out; while the field thins on its own, the stakes hold.

Deriving the level from that clock then meant an elimination reset the multiplier from four times face to one, which reads as the game forgetting what had just happened. It ratchets now: an elimination stops the climb without putting it back.

Escalation has to move both sides

In a live match a clue worth 2,000 paid the winner 500 and charged the loser the full 2,000 — the ceiling clipped the winnings while the loss went through untouched. The stakes only ever moved one way, and did it invisibly, which read as overtime not working at all. The winner of a raised clue now banks the whole amount and is clipped on the next clue instead, so the falling roof takes it back unless they keep winning.

A very long stall — eight escalation windows, about 48 clues with nobody eliminated — opens overtime even with people still queued, because a slow entry interval means the queue may never empty. That threshold has to be high: at three windows it fired during the entry phase and cost eleven minutes of match length and twenty points of draw fairness.

WHAT ELSE IS IN THERE

Details that matter in the room rather than on paper.

Wrestlers

Every player gets an 8-bit wrestler: singlet style and colour, hair style and colour, skin tone, chosen on arrival to be distinguishable from everybody already in the room. Thirty players get thirty tellable-apart wrestlers. One module draws both the portrait on a score tile and the figures in the animations, so a player is recognisably themselves being thrown out of the ring — and when nobody wins the clue, a referee in stripes does the throwing.

The portrait is drawn for the job rather than cropped from the body sprite. Cropping was the obvious approach and it was wrong: the body's head is four pixels across, so at tile size every player was an identical skin-coloured blob.

Turning up late, and who is next

A latecomer joins at any point, goes to the back of the queue and enters on the standard stake. With the interval on auto it recomputes against the clues actually left, so entries compress as the roster grows. It is refused once overtime has opened — letting somebody in then would reopen it while the ratchet held the stakes high.

Who is coming next is hidden from the room by default. The countdown stays, because knowing when somebody arrives is tactical; only the name goes, so the horn means something. The host sees it in the answers window, and the entrant always sees their own countdown.

Saves and gifts

Anyone still in the ring can put money up to buy an eliminated player back, declared with S and settled at the next clue so nothing pauses. Several people can chip in for one player, and partial amounts are allowed — a cheap save is a weak save. From the queue, a player waiting to come in can hand part of their entry to anybody in the ring and walks in lighter by exactly that much.

The watch screen

A public, read-only view of the board at /watch/CODE — no key, no login, and the answers are never sent to it. Meant to replace sharing a screen: a state push per event against a continuous video stream is a rounding error, and every viewer gets a crisp local render at their own resolution rather than a compressed copy of the host's monitor. One screen in the room can turn on the sound.

The entry countdown

Each waiting player sees their own entry, not the next entry in the match, with their place in the queue beside it. Three rising tones over their last three clues, and the page washes amber as they come up.

Match logs

Every match is recorded and saved three ways: on the server, in the host's downloads automatically, and on demand from the champion screen. Every clue, every buzz time, the connection it arrived over, scores before and after, entries, eliminations, corrections, and the predicted length against the real one. Nearly everything in this document that is a number rather than an opinion came out of one of those files.

METHOD

The rules engine is a plain module with no framework and no network, driven by a Monte Carlo harness that plays hundreds of matches per configuration using a simple model of player ability. Every figure in this document comes from that harness or from a recorded match.

The harness runs before every deployment. It has already caught a scoring change that pushed non-pot matches from 57 minutes to 746 — a regression that was invisible in ordinary play and would have been found on a Saturday night otherwise.

Clue material comes from a public dataset rather than from j-archive directly. The archive's maintainer asked not to be crawled, and a slow crawler is still a crawler.